

# Milestone 1.2: Maximum Likelihood

INFO 521 · Machine Learning Foundations · project milestone brief ·  
One Problem, Three Lenses · Lens 2 of 3

## How to read this

This is the assignment brief for one project milestone. The Satisfactory criteria below are authoritative: every one must be met, and it is all-or-nothing rather than partial credit. A Not-Yet returns with targeted feedback and one revise cycle. Do the work in the notebook distributed through Classroom 50; this PDF is the brief, not the workspace.

**Course-schedule placement.** Part 1 of the project: due Week 5, the midpoint. This milestone covers M3 and is due Week 4. Gated by Checkpoint quiz 1.2 (MLE for the weights, Wk 3).

## The second lens

In 1.1 you minimized squared error. Now you will **model the noise explicitly**. Assume each target is generated as

$$y_n = \mathbf{w}^\top \phi(x_n) + \epsilon_n, \quad \epsilon_n \sim \mathcal{N}(0, \sigma^2),$$

and estimate  $\mathbf{w}$  and  $\sigma^2$  by maximizing the likelihood. The payoff: you get *uncertainty*, not just a point prediction, and you will see that the MLE for  $\mathbf{w}$  is the very same answer least squares gave you.

Same ground rules: from-scratch NumPy/SciPy, no off-the-shelf solvers. Your own homework code is not “off the shelf”: Unit 3’s `w_mle` and `sigma2_mle` are exactly the estimators this milestone applies, and carrying them in is encouraged.

## What “Satisfactory” means for 1.2

- ☐ You **state the generative model** in prose and write the dataset log-likelihood under conditional independence.
- ☐ You implement the **MLE for  $\mathbf{w}$**  and show *numerically on your own data* that it equals your 1.1 OLS fit (this equivalence is the point).
- ☐ You estimate  $\hat{\sigma}^2$ .
- ☐ You compute and plot **predictive error bars**  $\hat{y} \pm 2\hat{\sigma}$ .
- ☐ **Play artifact**: error bars visibly widen where data is sparse / at the age range edges, with a written explanation.
- ☐ **Checkpoint quiz 1.2 (MLE for  $\mathbf{w}$ )** passed.

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## Setup

```
import numpy as np
from info521 import data, plotting, checks
plotting.use_course_style()
ds = data.load_clinical()
age = data.primary_predictor(ds)
y = ds["y"]
```

## 1 • The generative model and the likelihood

In a markdown cell, write the likelihood of the full dataset given i.i.d. Gaussian noise, and its log. Identify what maximizing it over  $w$  reduces to.

## 2 • MLE for the weights: and the equivalence

```
def mle_weights(X, y):
    # TODO: the analytic MLE for w under Gaussian noise. Your Unit 3 w_mle is
    # this function: bring it in (it was certified against the same fixture).
    raise NotImplementedError

# Reuse your 1.1 design_matrix / ols_fit (copy them in or import your module).
# Show numerically that mle_weights(X, y) == ols_fit(X, y) to tolerance, and
# say in one sentence WHY they coincide.
```

## 3 • MLE for the noise variance

```
def mle_sigma2(X, y, w):
    # TODO: (1/N) * sum of squared residuals. Your Unit 3 sigma2_mle is this
    # function: bring it in.
    # Self-check on the fixture:
    # checks.expect_close(mle_sigma2(checks.LINEAR_FIXTURE.X,
    #                                 checks.LINEAR_FIXTURE.y,
    #                                 checks.LINEAR_FIXTURE.w_ols),
    #                     checks.LINEAR_FIXTURE.sigma2_mle, "sigma^2 MLE")
    raise NotImplementedError
```

## 4 • Predictive uncertainty

Plot the fit over the data with a  $\pm 2\hat{\sigma}$  band using `plotting.error_band(...)`.

```
# TODO: grid of ages, predicted mean, constant-sigma band.
```

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### Play artifact: where is the model unsure?

Prediction uncertainty is largest at the **high-leverage extremes at both ends of the age range**, where a polynomial fit has the least surrounding data to pin it down, so it extrapolates rather than interpolates. Examine how prediction uncertainty behaves across the age range. (Hint: a constant  $\hat{\sigma}$  band is only the start, consider what *parameter* uncertainty would add at those extremes, which you will formalize in 1.3.)

**Write 150–250 words:** where is the model least trustworthy, and why? Connect this to the amount and location of the data.

```
# TODO: visualize and discuss.
```

### i Gate

Not Satisfactory until **Checkpoint quiz 1.2 (derive the MLE for  $w$ )** is passed. The checkpoint is an open-book D2L quiz, due the week before this milestone; the auto-graded portion resolves on submission and written responses allow one revision.

## Looking ahead

In 1.3 you put a **prior** on  $w$ . The point estimate becomes a posterior *distribution*, and the error bars you drew here gain a principled, data-location-aware shape.